That is

$$f(x)=\binom{n}{x}\left(\frac{1}{2}\right)^{n}, \qquad x=0,1,2,...,n.$$

The mean and variance are respectively are $\frac{n}{2}$ and $\frac{n}{4}$

Example 11.19

Find the binomial distribution for each of the following.

(i) Five fair coins are tossed once and X denotes the number of heads.

(ii) A fair die is rolled 10 times and X denotes the number of times 4 appeared.

Solution

(i) Given that five fair coins are tossed once. Since the coins are fair coins the probability of getting an head in a single coin is $p=\frac{1}{2}$ and $q=1-p=\frac{1}{2}$

Let X denote the number of heads that appear in five coins. X is a binomial random variable that takes on the values 0, 1,2,3,4 and 5, with $n=5$ and $p=\frac{1}{2}$.That is $X\sim B\left(5,\frac{1}{2}\right)$.

Therefore the binomial distribution is

$$f(x)=\binom{n}{x}p^{x}(1-p)^{n-x}, \qquad x=0,1,2,...,n$$

becomes

$$f(x)=\binom{5}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{n-x}, \qquad x=0,1,2,...,5.$$

That is

$$f(x)=\binom{5}{x}\left(\frac{1}{2}\right)^{n}, \qquad x=0,1,2,...,n$$

(ii) A fair die is rolled ten times and X denotes the number of times 4 appeared. X is binomial random variable that takes on the values $0, 1, 2, 3, \ldots 10$, with $n=10$ and $p=\frac{1}{6}$. That is $X\sim B\left(10,\frac{1}{6}\right)$.

Probability of getting a four in a die is $p=\frac{1}{6}$ and $q=1-p=\frac{5}{6}$.

Therefore the binomial distribution is

$$f(x)=\binom{10}{x}\left(\frac{1}{6}\right)^{x}\left(\frac{5}{6}\right)^{10-x}, \qquad x=0,1,2,...,10.$$

Example 11.20

A multiple choice examination has ten questions, each question has four distractors with exactly one correct answer. Suppose a student answers by guessing and if X denotes the number of correct answers, find (i) binomial distribution (ii) probability that the student will get seven correct answers (iii) the probability of getting at least one correct answer.

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 215 16-12-2022 16:52:30